

**Scheduled to Arrive, Predicted to Be Delayed: Using Machine Learning and Pre-Departure
Data to Anticipate U.S. Flight Delays**

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A flight may be scheduled to arrive on time, but conditions surrounding it can make the outcome difficult to predict. Flight Delays are a common problem for businesses, passengers, airports, and airlines, disrupting both personal and business plans. Delays may also affect airlines and airports because aircraft crews and gate operations are coordinated with connecting flight schedules. When one flight is delayed, the disruption may continue to affect later flights across multiple airports, creating a domino effect in which one flight delays the next. The ability to identify higher-risk flights before departure could therefore provide airports and airlines a way to handle these problems in advance, saving them a substantial amount of unforeseen cost annually.

Predicting flight delays is difficult because delays can result from several interacting conditions, especially when prediction is limited to information available before takeoff. Variables such as actual departure delays and reported delays are unknown to passengers as they prepare to board. Using them would also make a model appear more successful than it would be in a real predeparture setting. Therefore, this study relied on scheduled flight information, including the airline, date, scheduled departure and arrival times, duration, and distance.

While many studies have approached flight delays in various ways, including more complex modeling approaches like Dai (2024), which used a wider range of delay indicators, applied feature selection, clustering, and optimized a random forest model to predict flight delays. This study focused on predicting delays using predeparture data, specifically on predicting whether a completed non-diverted US domestic flight would arrive at least 15 minutes late, using only information available before departure. This allowed the project to evaluate how well scheduled flight information alone could anticipate a significant arrival delay.

Problem Statement

The purpose of this study was to compare machine learning models for predicting whether completed, non-diverted U.S. domestic flights would arrive at least 15 minutes after their scheduled arrival time using only information before flights departed. The study addressed the following research questions:

1. How effectively can significant arrival delays be predicted using only pre-departure information?
2. Which machine learning model performs best at identifying significantly delayed flights?
3. Do more complex models outperform simpler models when evaluated on future-month flight data?

Recall and F1 score were emphasized over accuracy since most flights in the dataset were not significantly delayed. A model could achieve relatively high accuracy by predicting that almost every flight is not delayed, even if it fails to identify delayed flights.

Data Description

The project used the 2015 Flight Delays and Cancellations dataset from Kaggle. The data included three primary files: flights, airports, and airlines. The flights file contained 5,819,079 observations and 31 variables, which included airports, airline, flight number, origin and destination airports, scheduled and actual departure times, flight distance, cancellation status, diversion status, arrival delay, and reported causes of delay. The dataset covered flights from January 1st through December 31st, 2015. The airport file contained 322 airport records and seven variables, including information on airport location and identification. The airlines file contained 14 airline codes and their corresponding airline names. A fourth file from the U.S. Department of Transportation's Bureau of Statistics was later used in data cleaning and provided

information on airport identifiers. It was needed because October records used a different airport identification format than the other months in 2015.

Flights that arrived 15 minutes or later were considered significantly delayed. Canceled and diverted flights were excluded because they were either outliers or lacked a standard arrival-delay outcome. After these exclusions, a total of 5,714,008 completed non-diverted flights remained in the dataset. Of these flights, 1,063,439, or 18.61%, were significantly delayed. The remaining 4,650,569 flights, or 81.39%, were not significantly delayed.

Exploratory Data Analysis

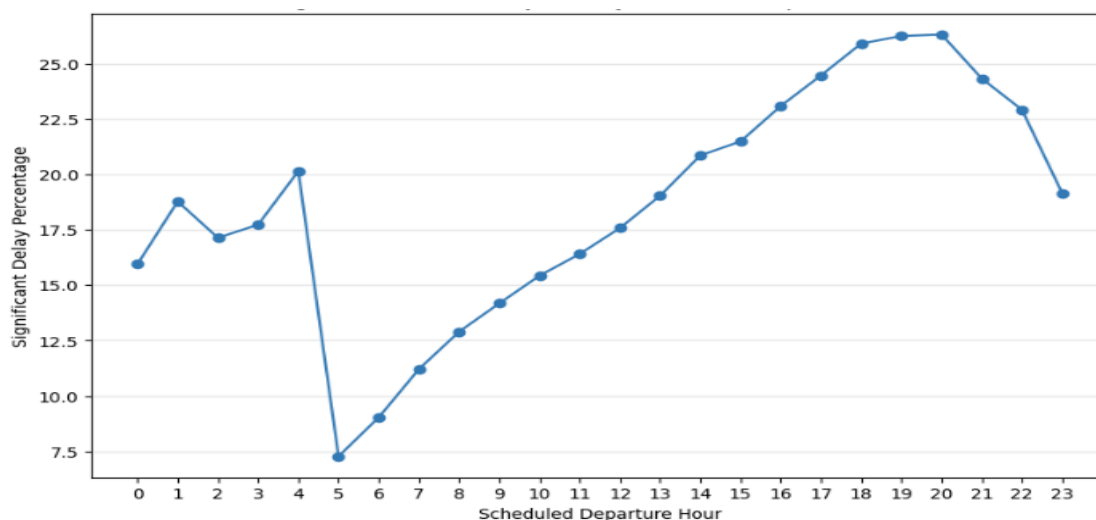
Exploratory data analysis was conducted to understand the data's structure, quality, and major patterns before modeling. The analysis included missing-value checks, duplicate checks, airport-code comparisons, cancellation and diversion counts, arrival-delay distributions, and delay rates across multiple flight characteristics. Arrival delay information was missing in 105,071 records, and further examination showed that these records comprised 89,884 canceled flights and 15,187 diverted flights, which were excluded from the modeling population because they lacked a standard arrival-delay outcome.

An important data quality issue was identified in October airport identifiers. Flights in most months used three-character airport codes such as LAX or DFW. However, all 486,165 October flight records used a five-digit numeric airport identifier. This resulted in 306 unmatched origin identifiers and 307 unmatched destination identifiers before cleaning. Delay patterns were also examined by month, airline, day of the week, scheduled departure hour, scheduled arrival hour, scheduled duration, distance, origin airport, destination airport, and route. Monthly significantly delayed flights varied throughout the year, ranging from about 12% to about 24%, with the highest ratings being January (21.00%), February (23.35%), and June (23.48%).

One of the clearest patterns observed involved departure time. As shown in Figure 1, flights scheduled in the morning generally had significantly lower delay rates than those scheduled in the afternoon, evening, and late-night hours. For example, flights scheduled to arrive around 7:00 am had a significant delay rate of nearly 9%, while late-evening flights had a delay rate of between approximately 20-25%.

Figure 1

Significant Arrival Delay Rate by Scheduled Departure Hour



Note. Flights scheduled later in the day generally experience higher delay percentages.

A correlation analysis of numeric pre-departure variables showed that correlations with the delay target were generally weak. The largest correlations were with the scheduled departure hour (approximately 0.14) and the scheduled arrival hour (approximately 0.12). The scheduled flight duration and distance showed an approximate correlation of 0.98, indicating the variables contained similar information. This suggested that significant delays were more likely associated with schedule, calendar, airline, duration, and distance characteristics than with a single dominant numerical predictor.

Data Preprocessing and Feature Engineering

The main data-quality issue involved October airport identifiers. Exploratory analysis found 306 unmatched origin identifiers and 307 unmatched destinations across 486,165 October records. While the other months used three-letter airport codes, October used five-digit U.S. Department of Transportation airport IDs. Rather than removing an entire month, the identifiers were converted using the BTS Master Coordinate table. The BTS reference was limited to airport records active on October 1, 2015; the conversion preserved valid three-letter codes and mapped the numeric IDs to their corresponding airport codes.

The modeling population was limited to completed, non-diverted flights with a recorded arrival delay and required pre-departure information. After exclusions, 5,714,008 eligible flights remained. SIGNIFICANT_DELAY was defined as 1 when ARRIVAL_DELAY was at least 15 minutes and 0 otherwise, resulting in a positive class rate of 18.61%. Missing values were handled based on whether a field was required and whether it would be available before departure. Post-departure and delay cause fields were not imputed because they were either outcome-related or unavailable at prediction time. TAIL_NUMBER was excluded because it contained 14,721 missing values and nearly 4900 unique aircraft identifiers. The six records with missing SCHEUDLED_TIME were removed because scheduled duration was required for feature engineering. Scheduled departure and arrival times were stored in HHMM format and were converted to minutes after midnight. The value 2400 was treated as midnight, and separate variables for scheduled departure and arrival hours were created.

All engineered features used information available before departure. ROUTE combined origin and destination; IS_WEEKEND identified Saturday and Sunday; DEPARTURE_PERIOD and ARRIVAL_PERIOD grouped scheduled hours into time of day categories; SCHEUDLED_OVERNIGHT identified schedules that crossed midnight; DISTANCE_GROUP

created mileage categories; and SPEED_PROXY_MPH divided distance by the scheduled duration. However, since speed proxy used scheduled rather than actual flight information, it did not introduce future information. ARRIVAL_DELAY was used to create the target and was then removed from the predictors. Actual departure and arrival times, departure delay, taxi and wheel times, elapsed and air times, cancellation information, and individual delay cause fields were also excluded because they occur during or after the flight and would introduce data leakage. The final data quality checks required positive distance and scheduled duration, valid three-letter airport codes and scheduled hours, a binary target, and no missing values.

For model-specific preprocessing, a Scikit-Learn ColumnTransformer was used so transformations learned only from the training data. The final modeling pipeline used 12 numeric variables and four categorical variables. Numeric fields underwent median imputation and standardization, while categorical fields received most-frequent imputation and one-hot encoding, with unknown categories being ignored. Origin, destination, and route were not included in the final pipeline because of their high cardinality and memory requirements.

Since the study was designed to predict delays of future flights from earlier data, a time-based split was prepared before model training. Flights from January through October were used for training; November was used for validation and model selection; and December was reserved for final testing. Because the full training set imposed memory and runtime limitations in Google Colab, a reproducible random sample of 500,000 January through October training records was generated with random_state=42, while the November and December datasets remained unchanged.

Modeling Methods

Four models were created and compared for predicting whether a flight would arrive 15

minutes or later: Logistic Regression, Random Forest, XGBoost, and a Multi-Layer Perceptron (MLP) Neural Network. A Logistic Regression was used as the baseline model for the project because it's a widely used method for binary classification. Logistic regressions also provides an interpretable benchmark. The Random Forest and XGBoost were included because they can capture more complex relationships in the data through ensemble learning. A Neural Network was also tested to examine whether a deep learning approach could improve predictive performance.

All four machine learning models used the same preprocessing pipeline and 500,000-record training samples described previously. Logistic regression used a maximum of 1,000 iterations and balanced class weights. XGBoost used 100 estimators, a maximum depth of 6, a learning rate of 0.1, and a positive class weight based on the ratio of non-delayed to delayed flights in training.

Since the target variable was imbalanced, with approximately 18.61% of flights classified as significantly delayed, the imbalance was addressed during model training to reduce bias. Logistic regression and random forest used balanced class weights, while XGBoost used a positive class weight of approximately 4.35. The Neural Network contained two hidden layers with 64 and 32 units, used ReLU activation, the Adam optimizer, early stopping, and batch-normalized sample weights. An initial unweighted neural network was also evaluated but predicted only the majority class; the weighted neural network was used in the final model comparison.

Validation and Testing

A time-based split was used to evaluate the models in a realistic prediction setting.

Flights from January through October were used for training, flights from November were used for validation and model selection, and flights from December were reserved for final testing. Using this split better reflects how the model would perform on future flight data and helps prevent data leakage.

Model performance was evaluated using Accuracy, Precision, Recall, F1-score, ROC-AUC, and PR-AUC. Greater emphasis was placed on Recall and F1-score because this study's primary goal was to identify flights likely to experience significant delays before departure. ROC-AUC and PR-AUC were also used to compare each model's overall classification performance across different decision thresholds.

Results

The four machine learning models were first compared using the November validation set to determine which model best met the project objective. As shown in Table 1 below, Logistic Regression achieved the highest Recall (0.363) and F1-score (0.269), indicating it identified a larger proportion of significantly delayed flights than the other models. Random Forest and the Neural Network achieved higher accuracy. Still, they detected relatively few delayed flights, while XGBoost achieved higher recall than the Random Forest and Neural Network but remained below that of logistic regression. Although Logistic Regression achieved the lowest accuracy and precision, it was more successful at identifying the delayed flights the study focuses on. Since the dataset was imbalanced, higher accuracy could be achieved by predicting flights as not delayed. Based on the November validation results, Logistic Regression was selected as the strongest model for identifying significantly delayed flights.

Table 1*Model Performance on November Validation Set*

Model	Accuracy	Precision	Recall	F1	ROC_AUC	PR_AUC
Logistic Regression	0.699	0.214	0.363	0.269	0.606	0.204
Random Forest	0.821	0.260	0.093	0.136	0.609	0.209
XGBoost	0.796	0.247	0.166	0.199	0.607	0.209
Neural Network (w)	0.825	0.234	0.064	0.100	0.560	0.184

After model comparison and selection using November data, performance was evaluated on the separate December test set to assess how well the selected model performed on future, unseen data. As shown in Table 2, Logistic regression again achieved the highest recall (0.269) and F1 score (0.258). The XGBoost achieved the highest ROC-AUC (.584) and PR-AUC (0.248), while the neural network achieved the highest accuracy (0.784) and precision (0.305). Although it had the highest accuracy, it also had the lowest recall (0.036). The December results therefore supported the November finding that logistic regression was the most useful model for identifying delayed flights even though the other model performed better on some individual metrics.

Table 2*Model Performance for the December Test Set*

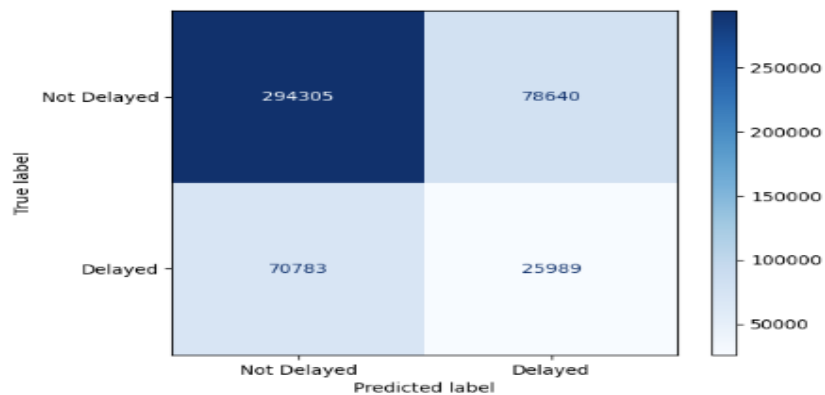
Model	Accuracy	Precision	Recall	F1	ROC_AUC	PR_AUC
Logistic Regression	0.682	0.248	0.269	0.258	0.575	0.242
Random Forest	0.761	0.260	0.086	0.130	0.579	0.246
XGBoost	0.721	0.249	0.176	0.206	0.584	0.248
Neural Network (w)	0.784	0.305	0.036	0.064	0.558	0.242

Since the logistic regression was selected as the final model, its December predictions were examined separately and in more detail. As shown in Figure 2, the model produced 25,989 true positives, 70,783 false negatives, 78,640 false positives, and 294,305 true negatives.

Although the model still missed many delayed flights, it did identify more delayed flights than the other models and achieved the strongest balance between recall and F1 score.

Figure 2

Logistic Regression Confusion Matrix for December Test Set



Final Model Selection

Although several models achieved higher overall accuracy than Logistic Regression, the project's primary objective was to identify flights that were likely to experience significant delays. For this reason, Recall and F1-score were considered more important than Accuracy when selecting the final model.

Logistic Regression achieved the highest Recall and F1-score among the evaluated models, indicating that it was more effective at detecting delayed flights while maintaining a reasonable balance between false positives and false negatives. Although the Random Forest, XGBoost, and Neural Network achieved higher accuracy or stronger performance on certain metrics, they identified fewer delayed flights. Based on these results, Logistic Regression was selected as the final model because it best aligned with the project's prediction goal.

Limitation and Future Work

This project has several limitations. Firstly, the dataset was highly imbalanced, with significantly fewer delayed flights than on-time flights. Although class imbalance was addressed prior to modeling, identifying these delayed flights remained challenging. Secondly, due to the memory limitations of the free version of Google Colab, model development was performed using a random sample of the 500,000 training data points after reducing the dimensionality of high-cardinality categorical features. While this approach allowed the models to be trained successfully, it limited performance; additional computing resources could enable training on the full dataset.

Future work that needs to be considered and improved includes adding additional hyperparameter tuning, testing other gradient boosting methods such as LightGBM or CatBoost, and exploring more advanced neural network architectures. Model performance may also improve by incorporating additional pre-departure information, such as weather conditions, airport congestion, or historical delay patterns. These are also important to consider because airlines have to take them into account and sometimes make slight adjustments to flight patterns, which may be highly correlated with delays.

Conclusion

This project developed and evaluated several machine learning models to predict whether a flight would arrive at least 15 minutes late using only information available before departure. Overall, the results show that pre-departure flight information can provide some useful signal for identifying significant delays, but prediction remained challenging. The findings also demonstrate that selecting a model based on the project's objective is more meaningful than relying solely on accuracy, especially when working with an imbalanced dataset.

References

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e-statistics](https://www.bts.gov/browse-statistical-products-and-data/statistical-products/airline-tim-e-statistics)

Appendix

Complete Project Source Code

Google Colab:

<https://colab.research.google.com/drive/1xFcLjedvqDG0U6cRcvzuTzVfOp0-vInA?usp=sharing>

Google Drive:

<https://drive.google.com/drive/folders/1N38b9B2Y5sB2QGI9R7MbGqrYKOGmiYOx?usp=sharing>

Data files may be accessed directly from the source or the Google Drive link above. The complete project source code is provided on the following pages and is also available via Google Colab.